

Gabriel Pitrella

Senior Software & Solutions Engineer | Industrial Engineer | Data, Cloud & Process Transformation

Industrial Engineering · Software Engineering · Data & AI · Manufacturing & Supply Chain

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PROFESSIONAL SUMMARY

Industrial Engineer with 7+ years of experience in Toyota Argentina's manufacturing and quality operations and 5+ years building production software and data systems for international clients. I combine process engineering, software development, cloud platforms, and data to turn complex operational requirements into scalable technical solutions.

At Toyota, I applied Kaizen and continuous-improvement methodologies to production and quality processes and built internal applications for operational data visualization. In software, I have led end-to-end implementations using TypeScript, Node.js, Next.js, Azure, AWS, SQL, Docker, Kubernetes, and AI-enabled architectures, working directly with international stakeholders.

My differentiator is the ability to move across the full problem space: understanding the business process, translating requirements into systems, implementing the solution, and measuring whether it actually improved the outcome.

DOMAIN EXPERTISE

Business & Systems Analysis: Requirements gathering and documentation, translating business processes into technical solutions, stakeholder management, cross-functional coordination, delivery and project coordination, process analysis and optimization

Industrial Engineering + Software: Digitalization of manufacturing and quality operations, internal tooling and operational data visualization for production environments, Kaizen and continuous improvement, structured problem-solving, ISO/TS 16949 and ISO 9001 environments

PROFESSIONAL EXPERIENCE

Senior Full Stack Engineer - Technical Solutions & Client Implementations | HubSnacks

Jul 2022 - Present

The Alt' HubSpot Agency. Technical lead and point of contact for international clients, translating business, operational, and data requirements into production systems.

- Own end-to-end delivery of client implementations, from requirements discovery and technical scoping through architecture, development, deployment, and optimization.
- Work directly with international stakeholders to translate business processes and data requirements into scalable technical solutions and measurable deliverables.
- Designed and implemented data-driven company and association directories integrating CRM data, enrichment logic, business rules, access control, and downstream reporting requirements.
- Delivered a multi-role (4 user profiles) institutional training platform for NATO, standalone from HubSpot: Azure SQL persistence, Azure Blob Storage, Keycloak/OIDC/OAuth authentication with role-based access, containerized on Docker/Kubernetes, backend migrated from Azure Functions to Node.js, Next.js frontend, coordinated development across multiple contributors.
- Built a card-style company/association directory within HubSpot CMS for NATO, fetching and displaying different HubSpot CRM objects and their associations.
- Maintained CI/CD pipelines and contributed to performance, reliability, and deployment improvements across client implementations.

Stack: TypeScript | Node.js | Next.js | Express | Azure SQL | Azure Blob | Azure Functions | HubSpot | Docker | Kubernetes

Quality & Process Assembly Engineer | Toyota Argentina

Nov 2014 - Mar 2022

- Managed 3 external suppliers as part of quality and production-support operations, coordinating deliverables and resolving quality issues cross-functionally.
- Redesigned line-station ergonomics and work-cycle timing as part of a Kaizen initiative, reducing the number of operators required per production line while improving worker ergonomics.
- Analyzed production and quality processes and translated operational requirements into data-driven improvement initiatives and decision-support tools.
- Built internal Next.js/AWS applications to visualize vehicle and production variables, giving production, quality, and engineering teams a shared operational view.
- Collaborated cross-functionally with production, quality, engineering, and other stakeholders to identify problems, define requirements, implement improvements, and evaluate results.

Quality & Project Engineer | Pedro Bucciero Industries

Sep 2009 - Oct 2014

- Strengthened quality control and inspection processes for automotive parts supply projects (PSA, GM, FIAT) under ISO/TS 16949:2009 and ISO 9001:2008 standards.
- Programmed CNC/PLC (G&M code) systems and coordinated cross-functional engineering teams across the production/supply pipeline.

SELECTED PROJECTS

Memxus - AI Memory & Retrieval System

2025 - Present

- Built and evaluated an AI memory/retrieval system using Supabase pgvector, OpenAI embeddings, MCP, Node/TypeScript and Python/FastAPI. Evaluated against LongMemEval-V2 (451 questions), measuring baseline, actual performance and theoretical ceiling: 2.66%, 20.18%, and 72.51%.
- Ran controlled ablation experiments to identify which system components improved retrieval performance, including reporting negative results.
- Exposed as an MCP server used by Claude, ChatGPT, and Cursor; submitted to the Anthropic Claude Connectors Directory, Slack App Marketplace, and Cursor Marketplace.

L'AQUA Store - E-commerce Platform

Feb 2021 - Sep 2023

- Designed and implemented a full e-commerce platform end-to-end with React and Next.js, including payment gateway integrations and analytics-driven optimization.
- Built and shipped independently, in parallel with the final years at Toyota Argentina.

CORE ENGINEERING

Languages: TypeScript, JavaScript, Python, SQL

Backend & APIs: Node.js, Express, REST, GraphQL, Serverless

Frontend: React, Next.js, Redux Toolkit

Cloud & Infrastructure: Azure, AWS, Docker, Kubernetes, CI/CD

Data: PostgreSQL, Azure SQL, Supabase, MongoDB, Data Integration, REST APIs

AI & Applied Systems: RAG, Vector Search, OpenAI Embeddings, AI Agents, MCP, OAuth 2.1/PKCE

EDUCATION

- Postgraduate, Industrial Automation | University of Buenos Aires | Apr 2009 - Mar 2011
- Industrial Engineer | National University of Cuyo | Feb 2002 - Aug 2008
- Full Stack Web Developer | Henry Bootcamp | Feb 2022 - Dec 2022

CERTIFICATIONS

- HubSpot Architecture I: Data Models and APIs | HubSpot Academy | Aug 2026
- HubSpot Data Integrations | HubSpot Academy | Jul 2026
- Integrating With HubSpot I: Foundations | HubSpot Academy | Jul 2026
- Advanced JavaScript | Udemy | Jun 2025
- Jira Project Management Specialization | LinkedIn Learning | Oct 2024

LANGUAGES

Spanish (Native) | English (C1 Advanced)